An Introduction to Sigma Models and Mirror Symmetry

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Abstract

This expository paper explains how sigma models organize a large part of the interaction between geometry and quantum field theory. Starting from supersymmetric quantum mechanics, we introduce the non-linear sigma model, the free boson on a circle, T-duality, and the basic algebra of two-dimensional $\mathcal{N} = (2, 2)$ supersymmetry. We then discuss supersymmetric non-linear sigma models with Kähler target, anomaly cancellation, topological A- and B-twists, Landau–Ginzburg models, and mirror symmetry. The quintic threefold is used as the central geometric example. Finally, we review stable maps, Gromov–Witten invariants, quantum cohomology, and Kontsevich's recursion for rational plane curves. The goal is to present a logically consistent mathematical account of the structures used in standard references on mirror symmetry and sigma models.

Contents

1 Introduction

A sigma model is a field theory whose bosonic fields are maps

$$\phi : \Sigma \longrightarrow X,$$

where Σ is a worldsheet or spacetime and X is a target manifold. In the simplest bosonic non-linear sigma model, a Riemannian metric g on X gives the Euclidean action

$$S(\phi) = \frac{1}{4\pi\alpha'} \int_\Sigma g_{ij}(\phi)\, \mathrm{d}\phi^i \wedge *\mathrm{d}\phi^j, \tag{1}$$

and possible additional terms may be built from a B-field, a potential, or fermionic partners. Thus a sigma model is not a model of "all quantum field theories", but rather a fundamental and flexible class of field theories in which the geometry of X is directly visible in the physics. This is why sigma models are central in string theory, mirror symmetry, and enumerative geometry [8, 4, 11].

The main organizing theme of this paper is the following: geometric data on the target can become algebraic or analytic data of the quantum theory, and conversely physical dualities can predict non-trivial mathematical equivalences. The most elementary instance is the free boson compactified on a circle, where the theory at radius R is equivalent to the theory at radius $1/R$ after exchanging momentum and winding. This is T-duality. In supersymmetric settings, T-duality becomes a prototype of mirror symmetry: it exchanges A-type and B-type structures and interchanges symplectic and complex-geometric information [4, 9].

For Calabi–Yau targets the relation becomes especially rich. The A-model depends only on the symplectic or complexified Kähler data, and its correlation functions are governed by Gromov–Witten invariants. The B-model depends only on the complex structure and is often more accessible by Hodge theory and variation of periods. Mirror symmetry predicts that the A-model of a Calabi–Yau manifold X is equivalent to the B-model of a mirror Y. Historically this led to striking enumerative predictions for rational curves on the quintic threefold [1, 2].

The paper is written as an expository final paper. I use path integrals in the standard physicist's sense, while indicating the mathematical objects that replace them in rigorous formulations: de Rham cohomology for supersymmetric quantum mechanics, stable maps and virtual fundamental classes for Gromov–Witten theory, and sheaf or Fukaya categories in the conjectural categorical refinement of mirror symmetry.

Conventions. Unless stated otherwise, $\alpha' = 1$ and all target manifolds are smooth. Complex coordinates on a Kähler manifold are written as $z^i, \bar{z}^{\bar{j}}$, and repeated indices are summed. The notation H is used for the hyperplane class on projective space.

2 Supersymmetric Quantum Mechanics

The first useful bridge between geometry and supersymmetry appears already in one-dimensional quantum mechanics. Let M be a compact Riemannian manifold with local coordinates x^i and metric g. The bosonic field is a path $x : S^1 \to M$. Its fermionic partners may be represented, after quantization, by exterior multiplication and contraction operators. The resulting Hilbert space is naturally identified with differential forms:

$$\mathcal{H} \cong \Omega^\bullet(M; \mathbb{C}). \tag{2}$$

Under this identification, the supercharges become

$$Q \leftrightarrow d, \qquad Q^\dagger \leftrightarrow d^\dagger, \qquad H = \frac{1}{2}\{Q, Q^\dagger\} \leftrightarrow \frac{1}{2}\Delta. \tag{3}$$

Therefore supersymmetric ground states are harmonic forms, and the space of ground states is canonically isomorphic to de Rham cohomology:

$$\mathcal{H}_0 \cong \operatorname{Ker} \Delta \cong H^\bullet_{\mathrm{dR}}(M; \mathbb{C}). \tag{4}$$

This is the basic model behind Witten's interpretation of Morse theory by supersymmetric quantum mechanics [10].

The Witten index is the graded trace

$$I_W(\beta) = \operatorname{Tr}_{\mathcal{H}}\left((-1)^F e^{-\beta H}\right), \tag{5}$$

where F is fermion number. Nonzero energy states come in boson–fermion pairs because Q and $Q^\dagger$ exchange the two parities. Hence the index is independent of β and equals the Euler characteristic:

$$I_W = \sum_k (-1)^k \dim H^k_{\mathrm{dR}}(M) = \chi(M). \tag{6}$$

This is the simplest instance of a general phenomenon: supersymmetric indices are often topological invariants.

A deformation by a Morse function $h : M \to \mathbb{R}$ replaces the differential by

$$Q_h = e^{-h} d e^h = d + dh \wedge . \tag{7}$$

The cohomology is unchanged, but in the semiclassical limit the path integral localizes near the critical points of h. If all critical points are nondegenerate, then

$$\chi(M) = \sum_{p \in \mathrm{Crit}(h)} (-1)^{\mu(p)}, \tag{8}$$

where $\mu(p)$ is the Morse index. This example foreshadows the localization principles used later in topological sigma models and Gromov–Witten theory.

Path integrals, traces, and the role of fermions

The equality between a Euclidean path integral on a circle and a trace in Hilbert space is a recurring theme. If H is the Hamiltonian of a quantum-mechanical system, then the Euclidean time evolution operator is $e^{-\beta H}$, and periodic boundary conditions on fields give

$$Z(\beta) = \int_{\text{periodic}} \mathcal{D}\phi \; e^{-S_E(\phi)} = \operatorname{Tr}_{\mathcal{H}}(e^{-\beta H}). \tag{9}$$

In a supersymmetric theory, periodic boundary conditions for both bosons and fermions compute instead

$$\mathrm{Tr}_{\mathcal{H}}\left((-1)^F e^{-\beta H}\right). \tag{10}$$

This insertion is responsible for the cancellation of nonzero energy states. It is also the reason the answer can often be computed by deforming the action to a limit in which only zero modes or classical solutions contribute.

For example, take $M = \mathbb{R}$ and a Morse function h. The supercharges may be represented formally as

$$Q = \psi^\dagger\big(p + ih'(x)\big), \qquad Q^\dagger = \psi\big(p - ih'(x)\big), \tag{11}$$

with $\{\psi, \psi^\dagger\} = 1$. Then

$$H = \frac{1}{2}p^2 + \frac{1}{2}(h'(x))^2 + \frac{1}{2}h''(x)[\psi^\dagger, \psi]. \tag{12}$$

When the potential confines the particle, the index is stable under rescaling $h \mapsto th$ and in the large t limit becomes a signed sum over critical points. This is the one-dimensional prototype of the stationary-phase and localization arguments used in higher-dimensional supersymmetric theories.

There is an analogous zero-dimensional Landau–Ginzburg toy model. Let $W : \mathbb{C}^n \to \mathbb{C}$ be holomorphic with isolated nondegenerate critical points. The Jacobi ring

$$\mathrm{Jac}(W) = \mathbb{C}[z_1, \dots, z_n] / \left(\frac{\partial W}{\partial z_1}, \dots, \frac{\partial W}{\partial z_n}\right) \tag{13}$$

will later appear as the ring of observables in the B-twisted Landau–Ginzburg model. Correlators localize at the critical locus, and in genus zero the residue formula has the form

$$\langle f_1 \cdots f_s \rangle_0 = \sum_{p \in \mathrm{Crit}(W)} \frac{f_1(p) \cdots f_s(p)}{\det(\partial_i \partial_j W)(p)}. \tag{14}$$

This is the Landau–Ginzburg counterpart of the localization behavior in supersymmetric quantum mechanics [4, 12].

A zero-dimensional toy model

It is helpful to begin even one dimension lower. In a zero-dimensional "field theory" the path integral is an ordinary integral. If x is a real bosonic variable and ψ_1, ψ_2 are Grassmann variables, consider

$$S(x, \psi_1, \psi_2) = \frac{1}{2}(h'(x))^2 - h''(x)\psi_1\psi_2, \tag{15}$$

where h is a smooth function with isolated nondegenerate critical points. Using the Berezin rule

$$\int d\psi_1 d\psi_2 \, \psi_1\psi_2 = 1, \tag{16}$$

one obtains

$$Z = \int dx\, d\psi_1 d\psi_2\, e^{-S} = \int dx\, h''(x) e^{-\frac{1}{2}(h'(x))^2}. \tag{17}$$

When the integral is interpreted with appropriate behavior at infinity, localization gives

$$Z = \sum_{h'(p)=0} \operatorname{sgn} h''(p), \tag{18}$$

the degree of the map $x \mapsto h'(x)$. The important point is not the particular normalization of the Gaussian, but the cancellation between bosonic and fermionic fluctuations. This cancellation is the finite-dimensional prototype of the Witten index and of the localization formulas that appear in topological sigma models.

There is a parallel complex version. Let $W : \mathbb{C} \to \mathbb{C}$ be holomorphic with isolated nondegenerate critical points and consider the zero-dimensional Landau–Ginzburg action

$$S = |\partial W|^2 - (\partial^2 W)\psi_1\psi_2 - \overline{\partial^2 W}\, \bar{\psi}_1 \bar{\psi}_2. \tag{19}$$

Integrating out the fermions cancels the Hessian determinant appearing in the stationary-phase expansion. Correlators of holomorphic functions localize at the critical points of W. Algebraically, the relevant ring is the Jacobi ring

$$\operatorname{Jac}(W) = \mathbb{C}[z]/(\partial W). \tag{20}$$

This elementary calculation explains why the Jacobi ring later appears as the B-model state space of a Landau–Ginzburg model.

The harmonic oscillator and Gaussian path integrals

The standard harmonic oscillator also clarifies the relation between path integrals and determinants. In Euclidean time, with period β, the action is

$$S_E[x] = \frac{1}{2}\int_0^\beta \left(\dot{x}^2 + x^2\right) dt = \frac{1}{2}\int_0^\beta x\left(-\frac{d^2}{dt^2} + 1\right) x\, dt. \tag{21}$$

The eigenfunctions of $-d^2/dt^2$ with periodic boundary condition have eigenvalues $(2\pi n/\beta)^2$, hence the formal Gaussian integral is proportional to

$$\det\left(-\frac{d^2}{dt^2} + 1\right)^{-1/2} = \prod_{n\in\mathbb{Z}} \left(1 + \frac{4\pi^2 n^2}{\beta^2}\right)^{-1/2}. \tag{22}$$

After zeta-function regularization this gives

$$Z(\beta) = \frac{1}{2\sinh(\beta/2)}, \tag{23}$$

which agrees with the operator calculation

$$\operatorname{Tr}(e^{-\beta H}) = \sum_{m\geq 0} e^{-\beta(m+1/2)}. \tag{24}$$

The same pattern recurs in the free boson: oscillator modes give eta-functions, zero modes give lattice sums, and supersymmetry modifies these determinants by cancellations between bosonic and fermionic modes. This is why the toy calculations above are not digressions but a useful preparation for sigma models.

Lessons for sigma models

The discussion so far suggests three principles that will be used repeatedly. First, the classical field space is usually infinite-dimensional, but supersymmetry often reduces protected quantities to finite-dimensional geometry. Second, fermions are not merely extra particles; after twisting or quantization they become the algebraic carriers of differential forms, polyvector fields, or obstruction theories. Third, the equality between a path integral on a closed worldsheet and a trace in a Hilbert space turns geometric invariance into algebraic identities such as associativity, index formulas, and recursion relations. These principles motivate the transition from quantum mechanics to two-dimensional sigma models.

3 The Free Boson on S^1 and T-duality

The most transparent sigma model duality is the compactified free boson. Let $X = S^1_R = \mathbb{R}/(2\pi R\mathbb{Z})$ and take the worldsheet to be a cylinder with coordinates (t,σ), $\sigma \sim \sigma + 2\pi$. A field $X(t,\sigma)$ satisfies

$$X(t,\sigma+2\pi) = X(t,\sigma) + 2\pi w R, \qquad w \in \mathbb{Z}. \tag{25}$$

The momentum along the target circle is quantized as n/R, $n \in \mathbb{Z}$. With $\alpha' = 1$, the left- and right-moving zero-mode momenta are

$$p_L = \frac{n}{R} + wR, \qquad p_R = \frac{n}{R} - wR. \tag{26}$$

The oscillator expansion decomposes the field into left- and right-moving parts, and the Hamiltonian is a sum of zero-mode and oscillator contributions.

Classical action and oscillator modes

The Lorentzian action of the free compact boson is

$$S = \frac{1}{4\pi}\int \mathrm{d}t\,\mathrm{d}\sigma\,\big((\partial_t X)^2 - (\partial_\sigma X)^2\big). \tag{27}$$

Its equation of motion is the wave equation

$$(\partial_t^2 - \partial_\sigma^2)X = 0, \tag{28}$$

so the general solution splits into left- and right-moving parts. In the sector of winding number w one writes schematically

$$X(t,\sigma) = x_0 + \frac{n}{R}t + wR\sigma + \text{oscillator modes}. \tag{29}$$

Canonical quantization gives creation and annihilation operators $\alpha_{-m}, \tilde{\alpha}_{-m}$ and $\alpha_m, \tilde{\alpha}_m$ for $m > 0$. The normal-ordering constant is the regularized zero-point energy of the oscillators; with the standard convention it contributes $-1/24$ to each chiral half. Therefore the chiral Hamiltonians are

$$L_0 - \frac{1}{24} = \frac{p_L^2}{4} + N - \frac{1}{24}, \qquad \bar{L}_0 - \frac{1}{24} = \frac{p_R^2}{4} + \tilde{N} - \frac{1}{24}. \tag{30}$$

The partition function (31) is precisely the trace of $q^{L_0-1/24}\bar{q}^{\bar{L}_0-1/24}$ over all momentum, winding, and oscillator states.

The torus partition function is

$$Z_R(\tau,\bar{\tau}) = \frac{1}{|\eta(\tau)|^2} \sum_{n,w\in\mathbb{Z}} q^{p_L^2/4}\bar{q}^{p_R^2/4}, \qquad q = e^{2\pi i\tau}. \tag{31}$$

Here $\eta(\tau)$ is the Dedekind eta function. This formula displays the equivalence

$$Z_R(\tau,\bar{\tau}) = Z_{1/R}(\tau,\bar{\tau}) \tag{32}$$

under the exchange

$$R \longleftrightarrow \frac{1}{R}, \qquad n \longleftrightarrow w. \tag{33}$$

At the level of fields, T-duality acts by changing the sign of the right-moving coordinate while keeping the left-moving coordinate fixed:

$$\widehat{X}_L = X_L, \qquad \widehat{X}_R = -X_R. \tag{34}$$

Consequently momentum and winding are exchanged. In the path-integral derivation one gauges the shift symmetry of the circle-valued field and introduces a Lagrange multiplier imposing flatness of the gauge field. Integrating out different variables gives the original radius R theory or the dual radius $1/R$ theory [8, 4].

More explicitly, in Euclidean signature the first-order action may be written in the form

$$S(B,\theta) = \frac{1}{4\pi}\int_\Sigma R^{-2} B \wedge *B + \frac{i}{2\pi}\int_\Sigma \theta\, \mathrm{d}B, \tag{35}$$

where B is a real one-form and θ is a periodic scalar. Integrating out θ imposes $\mathrm{d}B = 0$, so locally $B = R^2 \mathrm{d}X$ and one recovers the original radius-R model. Integrating out B instead gives a dual scalar θ with radius $1/R$. This derivation also shows why equations of motion and Bianchi identities are exchanged:

$$\mathrm{d} * \mathrm{d}X = 0 \quad \longleftrightarrow \quad \mathrm{d}\mathrm{d}\widehat{X} = 0. \tag{36}$$

The exchange is the field-theoretic origin of the momentum–winding interchange.

For a two-torus target, it is useful to combine the area and B-field into the complexified Kähler parameter

$$\rho = B + iA, \tag{37}$$

and to denote the complex structure by τ_T. T-duality along one circle exchanges these two parameters in the basic rectangular case. This is already a mirror phenomenon: complex and symplectic data are interchanged. The Strominger–Yau–Zaslow proposal generalizes this picture by regarding mirror symmetry for Calabi–Yau threefolds near the large complex structure limit as fiberwise T-duality on special Lagrangian torus fibrations [9].

The non-compact free boson and modular invariance

Before imposing the compact target condition, one may consider $X = \mathbb{R}$. The momentum p is continuous, and the oscillator trace gives the factor $|\eta(\tau)|^{-2}$. The zero-mode integration contributes a volume factor and a Gaussian integral,

$$\int_{-\infty}^{\infty} dp\, e^{-\pi\tau_2 p^2} = \frac{1}{\sqrt{\tau_2}}. \tag{38}$$

Consequently the non-compact free boson partition function has the schematic form

$$Z_{\mathbb{R}}(\tau,\bar\tau) = \frac{\mathrm{Vol}(\mathbb{R})}{2\pi\sqrt{\tau_2}}\,|\eta(\tau)|^{-2}. \tag{39}$$

This formula is not a finite number because the target is non-compact, but it illustrates a crucial point: modular covariance requires the zero-mode Gaussian and the oscillator eta-function to transform together. In the compact theory, the integral over continuous momentum is replaced by a Narain lattice sum, and modular invariance becomes a precise arithmetic property of an even self-dual lattice.

Compactification as a lattice theory

For the circle target, each sector is labeled by $(n,w) \in \mathbb{Z}^2$, and the pair (p_L, p_R) lies in the lattice

$$\Gamma_R^{1,1} = \left\{ \left(\frac{n}{R} + wR, \frac{n}{R} - wR \right) : n, w \in \mathbb{Z} \right\}. \tag{40}$$

The difference of squares is integral in the normalization used here:

$$\frac{p_L^2 - p_R^2}{2} = 2nw. \tag{41}$$

Level matching requires $L_0 - \bar L_0 \in \mathbb{Z}$, which is precisely compatible with this integrality. T-duality is the lattice isometry

$$(n,w;R) \longmapsto (w,n;1/R). \tag{42}$$

At the self-dual radius $R = 1$ the lattice has additional symmetry, and the corresponding conformal field theory has enhanced current algebra. This enhancement is a warning that geometry alone does not always display all symmetries of a quantum theory.

The torus target and the first appearance of mirror symmetry

Let the target be a real two-torus $T^2 = \mathbb{R}^2/\Lambda$ with metric G and constant B-field. The classical action contains

$$S = \frac{1}{4\pi}\int_\Sigma G_{ij}\, dX^i \wedge *dX^j + \frac{i}{2\pi}\int_\Sigma B_{ij}\, dX^i \wedge dX^j. \tag{43}$$

For a rectangular torus with radii R_1, R_2, one can package the geometric data into two complex parameters:

$$\sigma = i\frac{R_1}{R_2}, \qquad \rho = \frac{1}{2\pi}\int_{T^2} B + i\,\mathrm{Area}(T^2). \tag{44}$$

The parameter σ is the complex structure of the torus, while ρ is the complexified Kähler parameter. T-duality along exactly one circle exchanges them, up to the conventional action of $SL(2,\mathbb{Z})$ on each parameter. This is the most elementary mathematical model for the mirror slogan

$$\text{complex geometry on one side} \quad \leftrightarrow \quad \text{symplectic or Kähler geometry on the other side.} \tag{45}$$

In higher-dimensional Calabi–Yau geometry this exchange becomes vastly more subtle because instanton corrections and singular torus fibrations enter. Nevertheless, the torus computation is the conceptual seed of the SYZ picture.

4 Two-dimensional $\mathcal{N} = (2,2)$ Supersymmetry

Mirror symmetry is naturally formulated for two-dimensional $\mathcal{N} = (2,2)$ supersymmetric quantum field theories. Such a theory has four supercharges

$$Q_+, Q_-, \qquad \bar{Q}_+,\ \bar{Q}_-, \tag{46}$$

where the signs refer to chirality on the worldsheet. Ignoring central charges, the basic anticommutation relations are

$$Q_\pm^2 = \bar{Q}_\pm^2 = 0, \qquad \{Q_\pm, \bar{Q}_\pm\} = H \pm P, \qquad \{Q_+, Q_-\} = \{Q_+, \bar{Q}_-\} = 0, \tag{47}$$

with analogous relations obtained by conjugation. The theory also often carries two $U(1)$ R-symmetries, the vector and axial symmetries, with charges F_V and F_A. The mirror automorphism of the algebra is

$$Q_- \longleftrightarrow \bar{Q}_-, \qquad F_V \longleftrightarrow F_A, \tag{48}$$

while Q_+ and $\bar{Q}_+$ are left fixed. This algebraic fact is the reason mirror symmetry exchanges chiral and twisted chiral multiplets [4].

Superspace notation

Let $x^{\pm} = x^0 \pm x^1$. Superspace has odd coordinates $\theta^{\pm}, \bar{\theta}^{\pm}$. A convenient convention for the supercharges and covariant derivatives is

$$\mathcal{Q}_{\pm} = \frac{\partial}{\partial\theta^{\pm}} + i\bar{\theta}^{\pm}\partial_{\pm}, \qquad \bar{\mathcal{Q}}_{\pm} = -\frac{\partial}{\partial\bar{\theta}^{\pm}} - i\theta^{\pm}\partial_{\pm}, \tag{49}$$

$$D_{\pm} = \frac{\partial}{\partial\theta^{\pm}} - i\bar{\theta}^{\pm}\partial_{\pm}, \qquad \bar{D}_{\pm} = -\frac{\partial}{\partial\bar{\theta}^{\pm}} + i\theta^{\pm}\partial_{\pm}. \tag{50}$$

They satisfy

$$\{\mathcal{Q}_{\pm}, \bar{\mathcal{Q}}_{\pm}\} = -2i\partial_{\pm}, \qquad \{D_{\pm}, \bar{D}_{\pm}\} = 2i\partial_{\pm}, \tag{51}$$

and the D's anticommute with the $\mathcal{Q}$'s. This is why constraints such as $\bar{D}_{\pm}\Phi = 0$ are preserved by supersymmetry transformations.

A chiral superfield Φ satisfies $\bar{D}_{\pm}\Phi = 0$, while a twisted chiral superfield Y satisfies

$$\bar{D}_{+}Y = 0, \qquad D_{-}Y = 0. \tag{52}$$

The supersymmetric actions are built from three standard types of superspace integrals:

$$\text{D-terms:} \qquad \int \mathrm{d}^2x\, \mathrm{d}^4\theta\ K(\Phi, \bar{\Phi}), \tag{53}$$

$$\text{F-terms:} \qquad \int \mathrm{d}^2x\, \mathrm{d}\theta^{+}\mathrm{d}\theta^{-}\ W(\Phi) + \text{c.c.}, \tag{54}$$

$$\text{twisted F-terms:} \qquad \int \mathrm{d}^2x\, \mathrm{d}\theta^{+}\mathrm{d}\bar{\theta}^{-}\ \widetilde{W}(Y) + \text{c.c.}. \tag{55}$$

The D-term determines the target metric, an F-term determines an ordinary superpotential, and a twisted F-term appears naturally in gauge theories and T-dual descriptions. These distinctions are essential in the Landau–Ginzburg/Calabi–Yau correspondence and in mirror symmetry [12, 4].

In components, a single chiral multiplet contains a scalar ϕ, two Weyl fermions $\psi_{\pm}$, and an auxiliary field F. For flat target and superpotential W, the Lagrangian contains

$$\mathcal{L} = \partial_{\mu}\bar{\phi}\,\partial^{\mu}\phi + i\bar{\psi}_{-}(\partial_0 + \partial_1)\psi_{-} + i\bar{\psi}_{+}(\partial_0 - \partial_1)\psi_{+} + |F|^2 \tag{56}$$

$$+ F\partial_{\phi}W + \bar{F}\partial_{\bar{\phi}}\bar{W} - \partial_{\phi}^2 W\,\psi_{+}\psi_{-} - \partial_{\bar{\phi}}^2\bar{W}\,\bar{\psi}_{-}\bar{\psi}_{+}. \tag{57}$$

Eliminating F gives the scalar potential $|\partial_{\phi}W|^2$. Hence supersymmetric vacua of a massive Landau–Ginzburg model are the critical points of W, in agreement with the Jacobi-ring description.

Chiral and twisted chiral rings

The algebra in this section is not merely notation: it distinguishes two protected sectors. A local operator $\mathcal{O}$ is called chiral if

$$[\bar{Q}_{+}, \mathcal{O}] = [\bar{Q}_{-}, \mathcal{O}] = 0 \tag{58}$$

modulo operators that are $\bar{Q}_\pm$-exact. Twisted chiral operators satisfy the mixed condition

$$[\bar{Q}_+, \mathcal{O}] = [Q_-, \mathcal{O}] = 0. \tag{59}$$

Their equivalence classes form the chiral and twisted chiral rings. Mirror symmetry exchanges these rings, because it exchanges Q_- with $\bar{Q}_-$. In geometric phases this statement becomes the exchange of ordinary complex deformations with complexified Kähler deformations.

For a Landau–Ginzburg theory with chiral fields Φ_i and superpotential W, the chiral ring is

$$\mathbb{C}[\Phi_1, \ldots, \Phi_N]/(\partial_1 W, \ldots, \partial_N W). \tag{60}$$

In a non-linear sigma model on a Calabi–Yau manifold, the corresponding B-model ring is more naturally described by polyvector-valued cohomology. On the A-side the ring becomes quantum cohomology, where the classical cup product is corrected by holomorphic curves. These examples explain why the same supersymmetry algebra can lead simultaneously to algebraic geometry, symplectic geometry, and singularity theory.

A caution on terminology

The phrase "sigma model" is sometimes used broadly in physics, but in this paper it means a theory of maps from a worldsheet into a target space. This is narrower than the class of all quantum field theories. Gauge theories, for example, can flow in the infrared to sigma models on their vacuum moduli spaces, but they are not themselves sigma models in the same elementary sense. Witten's gauged linear sigma model is important precisely because it interpolates between a geometric non-linear sigma model phase and a Landau–Ginzburg phase [12]. Keeping this distinction avoids a logical gap in the original version of the paper.

5 Supersymmetric Non-linear Sigma Models and Anomalies

Let X be a Kähler manifold with local complex coordinates z^i and Kähler potential K. Then

$$g_{i\bar{j}} = \partial_i \partial_{\bar{j}} K \tag{61}$$

defines a Kähler metric. An $\mathcal{N} = (2, 2)$ non-linear sigma model with target X has bosonic field $\phi : \Sigma \to X$ and fermions

$$\psi_+ \in \Gamma(\Sigma, S_+ \otimes \phi^* T^{1,0} X), \quad \psi_- \in \Gamma(\Sigma, S_- \otimes \phi^* T^{1,0} X), \tag{62}$$

together with their complex conjugates. The bosonic part of the action is the energy functional (1), possibly plus a topological B-field term

$$S_B = \frac{i}{2\pi} \int_\Sigma \phi^* B. \tag{63}$$

The full supersymmetric action also contains fermion kinetic terms, curvature couplings, and auxiliary fields. Its local expression follows from the D-term with potential K [4].

A useful component form of the kinetic part is

$$\mathcal{L}_{\text{kin}} = g_{i\bar{j}}\partial_\mu\phi^i\partial^\mu\bar{\phi}^{\bar{j}} + ig_{i\bar{j}}\bar{\psi}_-^{\bar{j}}D_+\psi_-^i + ig_{i\bar{j}}\bar{\psi}_+^{\bar{j}}D_-\psi_+^i \tag{64}$$

$$+ R_{i\bar{j}k\bar{l}}\psi_+^i\psi_-^k\bar{\psi}_-^{\bar{j}}\bar{\psi}_+^{\bar{l}} + \text{auxiliary-field terms.} \tag{65}$$

The curvature term is not optional: it is forced by supersymmetry and by covariance under holomorphic coordinate changes on X. If a holomorphic superpotential $W : X \to \mathbb{C}$ exists, eliminating auxiliary fields adds a potential proportional to $\|\nabla W\|^2$ and Yukawa couplings involving the Hessian of W.

The axial and vector R-symmetries have different quantum behavior. The vector R-symmetry is non-anomalous for the ordinary Kähler sigma model. The axial R-symmetry has anomaly proportional to

$$\int_\Sigma \phi^* c_1(TX) = \langle c_1(TX), \phi_*[\Sigma]\rangle. \tag{66}$$

The anomaly can be seen from the fermion path-integral measure. The left- and right-moving fermions are sections of spin bundles tensored with $\phi^*T^{1,0}X$. The index of the relevant Dolbeault operator is

$$\text{index}(\bar{\partial}_{\phi^*TX}) = \int_\Sigma \phi^* c_1(TX) + (1-g)\dim_{\mathbb{C}} X. \tag{67}$$

The zero modes transform nontrivially under the axial symmetry, and the measure acquires a phase unless the c_1 contribution vanishes in every instanton sector. This is the precise correction to the common but misleading statement that every classical R-symmetry is automatically a quantum symmetry.

Thus the axial $U(1)_A$ symmetry survives for all instanton sectors precisely when $c_1(TX) = 0$. This is one reason Calabi–Yau manifolds are natural targets in two-dimensional $\mathcal{N} = (2,2)$ theories and in superstring compactification. More precisely, a compact Kähler manifold with $c_1(TX) = 0$ has a Ricci-flat Kähler metric in every Kähler class by Yau's theorem, and its holonomy is contained in $SU(n)$ under the usual additional hypotheses [5, 2].

Definition 5.1 (Calabi–Yau manifold)**.** *For the purposes of this paper, a Calabi–Yau n-fold is a compact Kähler manifold X with trivial canonical bundle $K_X \cong \mathcal{O}_X$. Equivalently, $c_1(TX) = 0$ in integral cohomology up to the usual torsion qualification. In many physics applications one also assumes X is simply connected and has holonomy exactly $SU(n)$.*

The anomaly statement also explains the existence of topological twists. The A-twist uses the vector R-symmetry and is available for any Kähler target. The B-twist uses the axial R-symmetry and therefore requires $c_1(TX) = 0$ for a non-anomalous quantum theory.

Why Calabi–Yau targets enter string compactification

The preceding anomaly argument is two-dimensional. It is related but not identical to the usual spacetime explanation for Calabi–Yau compactification in superstring theory. In the ten-dimensional superstring, one often considers a spacetime of the form

$$M^{3,1} \times X, \tag{68}$$

where $M^{3,1}$ is four-dimensional Minkowski spacetime and X is a compact real six-dimensional manifold. Preserving four-dimensional supersymmetry requires covariantly constant spinors on X, hence reduced holonomy. For a compact Kähler threefold, holonomy contained in $SU(3)$ is equivalent to the existence of a nowhere-vanishing holomorphic three-form and a Ricci-flat Kähler metric. Yau's theorem supplies such a metric when $c_1(X) = 0$ in a given Kähler class [5, 2].

Thus the Calabi–Yau condition appears from two complementary directions. From the worldsheet viewpoint it makes the axial R-symmetry non-anomalous and permits the B-twist. From the spacetime viewpoint it permits supersymmetric compactification. The agreement of these perspectives is one of the reasons the two-dimensional sigma model is such an efficient bridge between geometry and physics.

Dependence on geometric data

A useful summary is as follows. The classical bosonic sigma model depends on the metric g and B-field. Supersymmetry forces g to be Kähler in the $\mathcal{N} = (2, 2)$ case. The A-twisted theory remembers the symplectic form and B-field through the complexified class

$$B + i\omega \in H^2(X, \mathbb{C})/(H^2(X, \mathbb{Z}) \text{ shifts}), \tag{69}$$

but it forgets most metric details. The B-twisted theory remembers the complex structure and is insensitive to the Kähler class. Mirror symmetry is therefore not a mysterious equality between two arbitrary manifolds; it is a precise statement that two different geometric packages define equivalent protected sectors of quantum field theory.

6 Topological Twists

A topological twist changes the worldsheet spin by an R-charge. For an $\mathcal{N} = (2, 2)$ theory one obtains two basic twists:

$$\text{A-twist: use } U(1)_V, \qquad \text{B-twist: use } U(1)_A. \tag{70}$$

A compact way to remember the twists is the following table of spins after twisting. Here K_Σ denotes the canonical bundle of the worldsheet and $\phi^* T^{1,0} X$ is suppressed from the notation:

field	before twisting	A-twist	B-twist
ψ_+^i	S_+	$\bar{K}_\Sigma$	$\bar{K}_\Sigma$
ψ_-^i	S_-	$\mathcal{O}_\Sigma$	K_Σ
$\bar{\psi}_+^{\bar{i}}$	S_+	$\mathcal{O}_\Sigma$	$\mathcal{O}_\Sigma$
$\bar{\psi}_-^{\bar{i}}$	S_-	K_Σ	$\mathcal{O}_\Sigma$

The exact assignment depends on sign conventions, but the important point is invariant: after twisting, one linear combination of supercharges becomes a scalar BRST operator.

Each twist produces a scalar supercharge Q_{top} satisfying $Q_{\text{top}}^2 = 0$. Physical operators are then Q_{top}-cohomology classes. This is the mechanism that turns a supersymmetric field theory into a cohomological field theory [11, 4].

BRST cohomology and localization after twisting

After twisting, the scalar supercharge Q_{top} plays the role of a BRST operator. Operators differing by a Q_{top}-exact term are identified:

$$\mathcal{H}_{\text{phys}} = \operatorname{Ker} Q_{\text{top}} / \operatorname{Im} Q_{\text{top}}. \tag{71}$$

If the stress-energy tensor is Q_{top}-exact, then correlation functions of Q_{top}-closed operators do not change under deformations of the worldsheet metric. This is the sense in which the twisted theory is topological. The statement should be read carefully: the untwisted sigma model is a physical conformal field theory only under additional conditions, while the twisted model extracts protected observables that can often be defined mathematically.

Localization follows from adding a Q_{top}-exact positive term $t\{Q_{\text{top}}, V\}$ to the action. Since expectation values of closed operators are independent of t, one may take $t \to \infty$ and reduce the integral to the critical locus of the bosonic part of $\{Q, V\}$. In the A-model this critical locus is the space of holomorphic maps; in the B-model it is essentially the space of constant maps. This single difference explains why the A-model sees Gromov–Witten invariants whereas the B-model is controlled by complex-structure deformation theory.

The A-model

In the A-model, the local observables are naturally identified with de Rham cohomology classes of X:

$$\mathcal{O}_\alpha \quad \leftrightarrow \quad \alpha \in H^\bullet(X; \mathbb{C}). \tag{72}$$

Correlation functions localize on holomorphic maps $\phi : \Sigma \to X$. In genus zero, the three-point functions define the quantum product. More generally, for cohomology classes $\gamma_1, \ldots, \gamma_n$ and curve class $\beta \in H_2(X, \mathbb{Z})$, the corresponding Gromov–Witten invariant is

$$\langle \gamma_1, \ldots, \gamma_n \rangle_{g,\beta}^X = \int_{[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\text{vir}}} \prod_{i=1}^n \operatorname{ev}_i^*(\gamma_i). \tag{73}$$

The expected complex dimension of the moduli space of genus-g maps in class β is

$$\mathrm{vdim}_{\mathbb{C}}\,\overline{\mathcal{M}}_{g,n}(X,\beta) = (1-g)(\dim_{\mathbb{C}} X - 3) + n + \langle c_1(TX), \beta\rangle. \tag{74}$$

A correlator vanishes unless the total degree of insertions matches this virtual dimension. This selection rule is the A-model version of charge conservation, corrected by the axial anomaly.

Thus the A-model depends on the symplectic form and B-field through the factor $\exp(-\int_\beta(\omega - iB))$, but not on the complex structure except through choices used to define the theory.

A basic example is $X = \mathbb{P}^1$. If $H \in H^2(\mathbb{P}^1)$ is the hyperplane class, then the small quantum cohomology ring is

$$QH^\bullet(\mathbb{P}^1) \cong \mathbb{C}[H, q]/(H^2 - q). \tag{75}$$

Classically $H^2 = 0$, but quantum mechanically degree-one holomorphic spheres contribute the correction q.

The B-model

For a Calabi–Yau target, the B-model depends on the complex structure of X. Its local observables can be described by polyvector-valued Dolbeault cohomology

$$\bigoplus_{p,q} H^q(X, \wedge^p T^{1,0}X). \tag{76}$$

Using the holomorphic volume form of a Calabi–Yau manifold, this can be related to ordinary Hodge cohomology. For a Calabi–Yau threefold, the B-model deformation theory is governed by the Kodaira–Spencer class. Infinitesimal complex-structure deformations lie in

$$H^1(X, T^{1,0}X). \tag{77}$$

Contraction with a nowhere-vanishing holomorphic three-form identifies this group with $H^{2,1}(X)$. Thus $h^{2,1}$ counts complex-structure deformations, while $h^{1,1}$ counts Kähler deformations on the A-side. The exchange of these two Hodge numbers is the numerical core of Calabi–Yau mirror symmetry.

The B-model is not corrected by worldsheet instantons in the same way as the A-model, which is why mirror symmetry can convert difficult enumerative questions into period computations on the mirror [4, 2].

Landau–Ginzburg models

A Landau–Ginzburg model consists of a complex target, often $\mathbb{C}^n$, and a holomorphic superpotential W. In the B-twisted Landau–Ginzburg model with isolated critical points, the ring of local observables is the Jacobi ring

$$\mathrm{Jac}(W) = \mathbb{C}[x_1, \ldots, x_n]/(\partial_1 W, \ldots, \partial_n W), \tag{78}$$

and the genus-zero correlator is given by the residue formula

$$\langle f_1 \cdots f_s \rangle_0 = \sum_{p \in \mathrm{Crit}(W)} \frac{f_1(p) \cdots f_s(p)}{\det(\partial_i \partial_j W)(p)}. \tag{79}$$

For example, the mirror of the A-model on $\mathbb{P}^1$ can be represented by a Landau–Ginzburg model on $\mathbb{C}^*$ with

$$W(z) = z + qz^{-1}. \tag{80}$$

The critical equation $z - qz^{-1} = 0$ gives $z^2 = q$, matching the quantum relation $H^2 = q$ in (75). This small example captures the conceptual content of mirror symmetry in its simplest form [4, 12].

7 Gauged Linear Sigma Models

The preceding sections treated non-linear sigma models and Landau–Ginzburg models as separate constructions. Witten's gauged linear sigma model (GLSM) explains why they can be two phases of the same two-dimensional $\mathcal{N} = (2, 2)$ theory [12, 4]. This perspective is important for the academic logic of the paper: mirror symmetry is not simply an analogy between unrelated models, but often arises from moving in the parameter space of a single quantum field theory.

A basic GLSM consists of chiral fields Φ_i, a compact gauge group, a vector multiplet, a superpotential W, and a complexified Fayet–Iliopoulos parameter

$$t = r - i\theta. \tag{81}$$

The real parameter r controls a symplectic quotient, while θ becomes the B-field. For a $U(1)$ gauge theory with chiral fields of charges Q_i, the bosonic potential contains a D-term contribution of the form

$$U_D = \frac{e^2}{2} \left(\sum_i Q_i |\phi_i|^2 - r \right)^2. \tag{82}$$

In the low-energy limit $e \to \infty$, supersymmetric vacua must satisfy

$$\sum_i Q_i |\phi_i|^2 = r \tag{83}$$

modulo the gauge group. Thus the target space of the infrared sigma model is a symplectic quotient. In algebraic geometry this is the same mechanism as a GIT quotient, with the sign of r choosing a chamber.

Projective space from a GLSM

The simplest example has $N+1$ chiral fields $\Phi_0, \ldots, \Phi_N$, all of charge $+1$, and no superpotential. If $r > 0$, the D-term equation is

$$\sum_{i=0}^{N} |\phi_i|^2 = r. \tag{84}$$

Dividing by $U(1)$ gives $\mathbb{P}^N$. Hence the low-energy theory is the non-linear sigma model on projective space. The complexified parameter t becomes the complexified Kähler class of $\mathbb{P}^N$. This construction is the physical origin of the quantum parameter $q = e^{-t}$ in the relation

$$H^{N+1} = q. \tag{85}$$

In this sense the quantum cohomology of projective space is already visible in the gauge-theoretic presentation.

The quintic GLSM

The quintic threefold is obtained by taking a $U(1)$ GLSM with chiral fields

$$X_0, \ldots, X_4 \quad \text{of charge } +1, \qquad P \quad \text{of charge } -5, \tag{86}$$

and superpotential

$$W = P\, G_5(X_0, \ldots, X_4), \tag{87}$$

where G_5 is a homogeneous quintic polynomial. For $r \gg 0$, the D-term equation is

$$\sum_{i=0}^{4} |X_i|^2 - 5|P|^2 = r. \tag{88}$$

The F-term equations are

$$G_5(X) = 0, \qquad P\, \partial_i G_5(X) = 0. \tag{89}$$

For a smooth quintic, these force $P = 0$ and $G_5(X) = 0$. After quotienting by $U(1)$, the vacuum manifold is precisely the quintic hypersurface in $\mathbb{P}^4$. Thus the infrared theory is the non-linear sigma model on the quintic.

For $r \ll 0$, the field P cannot vanish. The gauge group is broken to the finite subgroup $\mathbb{Z}/5\mathbb{Z}$, and the remaining low-energy theory is a Landau–Ginzburg orbifold with superpotential $G_5(X)$. Therefore the geometric Calabi–Yau phase and the Landau–Ginzburg orbifold phase are connected inside a single GLSM parameter space. This statement is more precise than saying that the two theories are merely "similar"; they are two limiting descriptions of the same underlying model.

Anomaly cancellation in the GLSM

The same charges also explain the Calabi–Yau condition. For a $U(1)$ GLSM, the axial R-symmetry is non-anomalous when the sum of gauge charges vanishes:

$$\sum_i Q_i = 0. \tag{90}$$

For the quintic model,

$$1 + 1 + 1 + 1 + 1 - 5 = 0. \tag{91}$$

This is the gauge-theoretic counterpart of $c_1(TX) = 0$. More generally, a degree-d hypersurface in $\mathbb{P}^N$ is Calabi–Yau exactly when $d = N + 1$, which is the same condition that the GLSM charges sum to zero. The agreement among the adjunction formula, the worldsheet anomaly, and the GLSM charge condition is one of the cleanest ways to organize the logic of the subject.

Why this section matters for mirror symmetry

GLSMs give a systematic way to construct mirrors, especially for toric varieties and hypersurfaces in toric varieties. The Hori–Vafa mirror construction, for example, turns charged chiral fields into dual periodic variables and produces a Landau–Ginzburg mirror superpotential. Although a full derivation is beyond the scope of this final paper, the conceptual point is important: T-duality on circles becomes dualization of phases of charged fields; the result is a mirror Landau–Ginzburg model whose critical equations encode quantum cohomology. This is the technical bridge between the circle calculation of Section 3 and the quintic mirror calculation of Section 8.

8 The Quintic Threefold and Its Mirror

Hodge-theoretic formulation of the mirror relation

For a compact Kähler n-fold, the Hodge numbers are

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^q(X, \Omega^p_X). \tag{92}$$

They satisfy $h^{p,q} = h^{q,p}$ by complex conjugation and $h^{p,q} = h^{n-p,n-q}$ by Serre duality. For a mirror pair of Calabi–Yau n-folds one expects

$$h^{p,q}(X) = h^{n-p,q}(Y). \tag{93}$$

For $n = 3$, this means in particular

$$h^{1,1}(X) = h^{2,1}(Y), \qquad h^{2,1}(X) = h^{1,1}(Y). \tag{94}$$

This numerical test is necessary but not sufficient for mirror symmetry: many spaces can share Hodge numbers without defining equivalent conformal field theories.

The quintic threefold is the standard test case for mirror symmetry. Let

$$X = \{F_5(x_0, \ldots, x_4) = 0\} \subset \mathbb{P}^4 \tag{95}$$

be a smooth degree-five hypersurface. By the adjunction formula,

$$K_X \cong (K_{\mathbb{P}^4} \otimes \mathcal{O}_{\mathbb{P}^4}(5))|_X \cong \mathcal{O}_{\mathbb{P}^4}(-5+5)|_X \cong \mathcal{O}_X. \tag{96}$$

Thus X is Calabi–Yau in the sense of Definition 5.1. In terms of total Chern classes,

$$c(TX) = \frac{(1+H)^5}{1+5H}\bigg|_X = 1 + 10H^2 - 40H^3. \tag{97}$$

Therefore

$$\chi(X) = \int_X c_3(TX) = \int_X (-40H^3) = -40 \cdot 5 = -200. \tag{98}$$

The Lefschetz hyperplane theorem gives $h^{1,1}(X) = 1$. Since a Calabi–Yau threefold satisfies

$$\chi(X) = 2\big(h^{1,1}(X) - h^{2,1}(X)\big), \tag{99}$$

we obtain $h^{2,1}(X) = 101$. Equivalently, the complex moduli count is

$$\binom{5+4}{4} - 1 - \dim PGL(5) = 126 - 1 - 24 = 101. \tag{100}$$

The Hodge diamond is therefore

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 0 & & 0 & & \\ & 0 & & 1 & & 0 & \\ 1 & & 101 & & 101 & & 1 \\ & 0 & & 1 & & 0 & \\ & & 0 & & 0 & & \\ & & & 1 & & & \end{array}$$

The mirror quintic arises from the Greene–Plesser construction [3, 1]. Consider the Fermat pencil

$$X_\psi : \quad x_0^5 + x_1^5 + x_2^5 + x_3^5 + x_4^5 - 5\psi x_0 x_1 x_2 x_3 x_4 = 0. \tag{101}$$

It is invariant under diagonal transformations $x_i \mapsto \lambda_i x_i$, $\lambda_i^5 = 1$, subject to $\prod_i \lambda_i = 1$, modulo the diagonal subgroup already acting trivially on projective space. The effective group is isomorphic to $(\mathbb{Z}/5\mathbb{Z})^3$. The quotient X_ψ/G is singular, but a crepant resolution Y_ψ exists. Its Hodge numbers are exchanged:

$$h^{1,1}(Y_\psi) = 101, \qquad h^{2,1}(Y_\psi) = 1. \tag{102}$$

Thus X and Y have Hodge diamonds reflected across a diagonal. This reflection is a useful numerical shadow of mirror symmetry, but it is not the full statement: the deeper statement is the equivalence of the A-model on X with the B-model on Y, and conversely.

The famous enumerative consequence concerns rational curves on the quintic. The A-model numbers on X are hard because they count holomorphic curves. Mirror symmetry converts them into B-model period computations on Y. The calculations of Candelas et al. [1] predicted, for example, the number 2875 of lines on a generic quintic threefold and far more complicated higher-degree counts. Later mathematical foundations used Gromov–Witten theory and stable maps to make such invariants rigorous [7, 2].

Periods, the mirror map, and enumerative predictions

The practical power of the mirror quintic is that the B-model periods can be computed by solving a Picard–Fuchs differential equation. Set

$$z = (5\psi)^{-5}, \qquad \theta = z\frac{d}{dz}. \tag{103}$$

The periods of the holomorphic three-form on the mirror family satisfy the fourth-order equation

$$\left[\theta^4 - 5z(5\theta+1)(5\theta+2)(5\theta+3)(5\theta+4)\right]\Pi(z) = 0. \tag{104}$$

Near $z = 0$, the large complex structure limit, one chooses a basis of solutions with logarithmic behavior. The ratio of a logarithmic period to the holomorphic period defines the mirror coordinate

$$t = \frac{\Pi_1(z)}{\Pi_0(z)}, \qquad q = e^{2\pi i t}. \tag{105}$$

The B-model Yukawa coupling, expressed in the flat coordinate t, has the form

$$K_{ttt} = 5 + \sum_{d\geq 1} N_d \frac{d^3 q^d}{1 - q^d}. \tag{106}$$

Mirror symmetry interprets the integers N_d as genus-zero counts of degree-d rational curves on the original quintic. The first values are

$$N_1 = 2875, N_2 = 609250, N_3 = 317206375. \tag{107}$$

The value $N_1 = 2875$ was known classically as the number of lines on a general quintic threefold, while the higher values were among the striking predictions of mirror symmetry [1, 2]. In this form the relation between sigma models and enumerative geometry is very concrete: a period integral on one side determines curve counts on the other.

What the quintic example proves and what it does not prove

The quintic example is often presented as if Hodge-number exchange alone establishes mirror symmetry. That is not correct. The Hodge diamond calculation proves that the Greene–Plesser construction has the right numerical shape. The stronger claim is that the A-model of X and the B-model of Y have matching correlation functions after the mirror map. This stronger claim is supported by the period calculation above and by the subsequent mathematical theory of Gromov–Witten invariants, but it is not a consequence of the Hodge diamond alone. Separating these statements remedies a common logical gap in elementary expositions.

9 Gromov–Witten Invariants and Quantum Cohomology

The A-model requires a compactified moduli space of holomorphic curves. The appropriate object is the moduli space of stable maps introduced by Kontsevich.

Definition 9.1 (Stable map)**.** *Let X be a smooth projective variety. An n-pointed genus-g stable map to X in class $\beta \in H_2(X, \mathbb{Z})$ is a morphism*

$$f : (C, p_1, \ldots, p_n) \longrightarrow X$$

from a connected nodal curve such that $f_*[C] = \beta$ and the automorphism group of the data is finite. Equivalently, any contracted rational component must have at least three special points, and any contracted elliptic component must have at least one special point.

Virtual dimension and selection rules

The expected dimension formula has a simple deformation-theoretic origin. For a smooth map $f : C \to X$, infinitesimal deformations and obstructions are governed by

$$H^0(C, f^*TX) \quad \text{and} \quad H^1(C, f^*TX), \tag{108}$$

with corrections for deformations of the domain curve and marked points. Riemann–Roch gives

$$\chi(C, f^*TX) = \dim_{\mathbb{C}} X(1-g) + \langle c_1(TX), \beta\rangle. \tag{109}$$

Adding the dimension $3g - 3 + n$ of the moduli of pointed curves leads to

$$\mathrm{vdim}_{\mathbb{C}}\, \overline{\mathcal{M}}_{g,n}(X, \beta) = (1-g)(\dim_{\mathbb{C}} X - 3) + n + \langle c_1(TX), \beta\rangle. \tag{110}$$

This is the geometric version of the charge-selection rule in the A-model. For a Calabi–Yau threefold and genus zero it reduces to n, so three-point functions are dimensionally natural after inserting three cohomology classes whose degrees add appropriately.

The axioms behind quantum cohomology

Several formal properties of Gromov–Witten invariants are especially important. The axiom for the fundamental class says that inserting the identity often reduces the invariant to one with fewer marked points. The divisor axiom says that for a divisor D,

$$\langle D, \gamma_1, \ldots, \gamma_n\rangle_{0,\beta} = \left(\int_\beta D\right) \langle \gamma_1, \ldots, \gamma_n\rangle_{0,\beta} \tag{111}$$

when the usual stability conditions are satisfied. The splitting axiom describes what happens when a stable curve breaks into two components. This last axiom is the geometric reason for WDVV and for associativity of the quantum product.

In genus zero, choose coordinates t^a dual to a basis T_a of cohomology. The genus-zero potential is formally

$$\mathcal{F}_0(t) = \sum_{n,\beta} \frac{q^\beta}{n!} \langle t, \ldots, t \rangle_{0,n,\beta}, \tag{112}$$

where $t = \sum_a t^a T_a$. Its third derivatives are the structure constants of the quantum product. Associativity is equivalent to the WDVV equations

$$\sum_{e,f} \frac{\partial^3 \mathcal{F}_0}{\partial t^a \partial t^b \partial t^e} g^{ef} \frac{\partial^3 \mathcal{F}_0}{\partial t^f \partial t^c \partial t^d} = \sum_{e,f} \frac{\partial^3 \mathcal{F}_0}{\partial t^a \partial t^c \partial t^e} g^{ef} \frac{\partial^3 \mathcal{F}_0}{\partial t^f \partial t^b \partial t^d}. \tag{113}$$

This identity is the rigorous form of the physical statement that four-point functions factorize consistently in different channels.

The moduli stack is denoted

$$\overline{\mathcal{M}}_{g,n}(X, \beta). \tag{114}$$

It carries evaluation maps

$$\mathrm{ev}_i : \overline{\mathcal{M}}_{g,n}(X, \beta) \to X, \qquad (f : C \to X, p_1, \ldots, p_n) \mapsto f(p_i). \tag{115}$$

Because the moduli space may have components of unexpected dimension, Gromov–Witten invariants are defined using the virtual fundamental class:

$$\langle \gamma_1, \ldots, \gamma_n \rangle_{g,\beta}^X = \int_{[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}} \prod_{i=1}^n \mathrm{ev}_i^*(\gamma_i). \tag{116}$$

This is the mathematical counterpart of A-model correlation functions [7, 6, 2].

Quantum cohomology packages genus-zero three-point Gromov–Witten invariants into a product. Let $\{T_a\}$ be a homogeneous basis of $H^\bullet(X)$ and let $(g_{ab}) = (\int_X T_a \cup T_b)$ be the Poincare pairing, with inverse (g^{ab}). The small quantum product is defined by

$$T_a * T_b = \sum_{c,d} \sum_{\beta \in H_2(X,\mathbb{Z})} q^\beta \langle T_a, T_b, T_c \rangle_{0,\beta}^X \, g^{cd} T_d. \tag{117}$$

The WDVV equations, equivalently the associativity of this product, encode the factorization of boundary strata in the moduli space of stable maps.

Example 9.2 (Quantum cohomology of $\mathbb{P}^n$)**.** *For projective space, the small quantum cohomology ring is*

$$QH^\bullet(\mathbb{P}^n) \cong \mathbb{C}[H, q]/(H^{n+1} - q), \tag{118}$$

where H is the hyperplane class. This is the simplest higher-dimensional analogue of (75)*.*

For $X = \mathbb{P}^2$, let N_d denote the number of degree-d rational curves through $3d - 1$ general points. Equivalently,

$$N_d = \langle [pt], \ldots, [pt] \rangle_{0,d}^{\mathbb{P}^2}, \qquad \text{with } 3d - 1 \text{ insertions}. \tag{119}$$

Kontsevich's recursion determines all N_d from $N_1 = 1$:

$$N_d = \sum_{\substack{d_1+d_2=d \\ d_1,d_2>0}} \left[\binom{3d-4}{3d_1-2} d_1^2 d_2^2 - \binom{3d-4}{3d_1-1} d_1^3 d_2 \right] N_{d_1} N_{d_2}. \tag{120}$$

This formula can be derived from associativity of quantum cohomology, or geometrically from the boundary divisors in $\overline{\mathcal{M}}_{0,n}(\mathbb{P}^2, d)$ [7, 6]. The first few values determined by the recursion are

$$N_1 = 1, \quad N_2 = 1, \quad N_3 = 12, \quad N_4 = 620, \quad N_5 = 87304. \tag{121}$$

Geometrically, $N_3 = 12$ is the classical number of rational cubics through eight general points in the plane. From the sigma-model viewpoint these numbers are structure constants of the A-model, so their recursion is a consequence of associativity rather than an isolated combinatorial miracle.

How the recursion arises

Here is the idea behind (120). A degree-d rational plane curve passing through $3d-1$ points has expected dimension zero. To derive a recursion, one studies a one-dimensional family with $3d-2$ incidence conditions and then compares two boundary divisors in the moduli space of four-pointed rational curves. When the domain splits into two components of degrees d_1 and d_2, the marked points distribute between the two components. The binomial coefficients in (120) count these distributions, while the powers of d_1 and d_2 record the incidence multiplicities with lines and points. This is precisely the geometric content of the WDVV equation for $\mathbb{P}^2$.

The first terms illustrate the interpretation:

d	1	2	3	4	5
N_d	1	1	12	620	87304

A line is determined by two points, a conic by five points, and there are twelve rational cubics through eight general points. The values for $d \geq 4$ are already difficult by classical methods, but they follow uniformly from the quantum product. This example is intentionally placed after the quintic: it is a simpler surface case where the same A-model mechanisms can be written explicitly.

It illustrates why sigma models are mathematically powerful: the consistency of a quantum product imposes recursion relations on classical enumerative problems.

10 Mirror Symmetry in Summary

The preceding sections can be summarized by the following dictionary:

A-model on X	B-model on mirror Y
symplectic and complexified Kähler data	complex structure data
Gromov–Witten invariants	periods and variation of Hodge structure
quantum cohomology	complex-structure deformation theory
Lagrangian branes	coherent sheaves

This table should be read as a guide rather than a theorem in full generality. Different versions of mirror symmetry have different hypotheses and levels of rigor. The Hodge number identity $h^{p,q}(X) = h^{n-p,q}(Y)$ is only the coarsest numerical shadow of the phenomenon.

Two influential refinements are worth mentioning. First, Kontsevich's homological mirror symmetry predicts an equivalence between the Fukaya category of a Calabi–Yau manifold and the derived category of coherent sheaves on its mirror. Second, the Strominger–Yau–Zaslow conjecture predicts that, near a large complex structure limit, mirror symmetry should arise from dual special Lagrangian torus fibrations [9, 4].

Conjecture 10.1 (Homological mirror symmetry)**.** *For a mirror pair of Calabi–Yau manifolds (X, Y), the derived Fukaya category of X should be equivalent to the bounded derived category of coherent sheaves on Y, with the mirror statement obtained by interchanging X and Y.*

Conjecture 10.2 (SYZ mirror symmetry)**.** *Near a large complex structure limit, mirror Calabi–Yau manifolds should admit dual special Lagrangian torus fibrations over a common base. The mirror transformation is expected to be fiberwise T-duality, corrected by instanton effects.*

The mathematical power of sigma models comes from this chain of ideas: supersymmetry produces cohomological structures; topological twisting turns correlation functions into invariants; dualities identify different geometric descriptions of the same theory; and consistency conditions such as associativity become non-trivial theorems or conjectures in enumerative geometry. The subject remains active because no single framework captures all these structures simultaneously, but the examples above show why sigma models form a natural bridge between geometry and quantum field theory.

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